

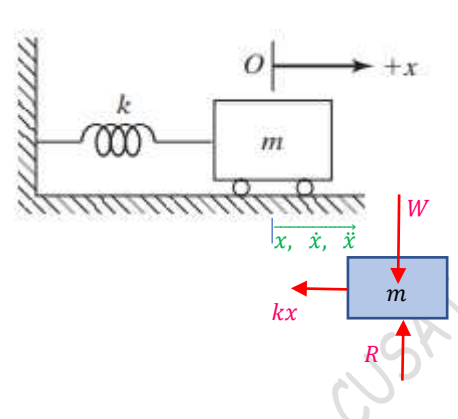
Undamped free vibration of s.d.o.f. systems

Consider a single degree of freedom mathematical model of a system as shown in the figure. Assume that no damping is present in the system. Let this system is disturbed from equilibrium and allowed to vibrate freely.

Consider an instant when the mass moves x distance to the right. The various forces acting on the system are marked in the free-body diagram. Writing the equation of motion in the x -direction,

$$m\ddot{x} = \Sigma F_x = -kx$$

$$m\ddot{x} + kx = 0 \dots \dots \dots (1)$$



This is the differential equation governing the free vibration of undamped single degree of freedom systems. This is a **second order**, **homogeneous**, **linear**, **ordinary differential equation** with **constant coefficients**. Any exponential function can be considered as its solution. Let it be,

$$x(t) = Ae^{pt} \dots \dots \dots (2)$$

Where A is an arbitrary constant and p is a constant (not arbitrary). Substituting (2) back in (1), we get,

$$m p^2 A e^{pt} + k A e^{pt} = 0 \quad \ll \gg \quad A e^{pt} (m p^2 + k) = 0 \quad \ll \gg \quad m p^2 + k = 0$$

$$p_{1,2} = \pm i \sqrt{\frac{k}{m}} = \pm i \omega_n$$

Now, we can have two solutions based on p_1 and p_2 as $x_1(t) = A_1 e^{p_1 t}$ and $x_2(t) = A_2 e^{p_2 t}$. Since the differential equation is of second order, the solution should contain two arbitrary constants. We can apply the principle of superposition as the vibration is linear.

$$x(t) = A_1 e^{p_1 t} + A_2 e^{p_2 t} = A_1 e^{i \omega_n t} + A_2 e^{-i \omega_n t}$$

The arbitrary constants can be evaluated from the initial conditions. But,

$$e^{i \omega_n t} = \cos \omega_n t + i \sin \omega_n t \quad \text{and} \quad e^{-i \omega_n t} = \cos \omega_n t - i \sin \omega_n t$$

$$x(t) = A_1 (\cos \omega_n t + i \sin \omega_n t) + A_2 (\cos \omega_n t - i \sin \omega_n t) = C_1 \cos \omega_n t + C_2 \sin \omega_n t$$

Let $C_1 = X_0 \sin \varphi$ and $C_2 = X_0 \cos \varphi$ so that

$$X_0 = \sqrt{C_1^2 + C_2^2} \quad \text{and} \quad \tan \varphi = \frac{C_1}{C_2}$$

$$x(t) = C_1 \cos \omega_n t + C_2 \sin \omega_n t$$

$$x(t) = X_0 \sin \varphi \cos \omega_n t + X_0 \cos \varphi \sin \omega_n t$$

$$x(t) = X_0 \sin(\omega_n t + \varphi)$$

The arbitrary constants A_1 and A_2 or C_1 and C_2 or X_0 and φ need to be evaluated from the initial conditions.

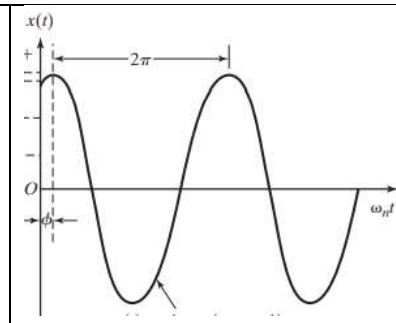
Let the initial displacement be x_0 and initial velocity $\dot{x}_0$

$$x(t) = C_1 \cos \omega_n t + C_2 \sin \omega_n t$$

$$\dot{x}(t) = -C_1 \omega_n \sin \omega_n t + C_2 \omega_n \cos \omega_n t$$

When $t = 0$, $x = x_0$, $\dot{x} = \dot{x}_0$

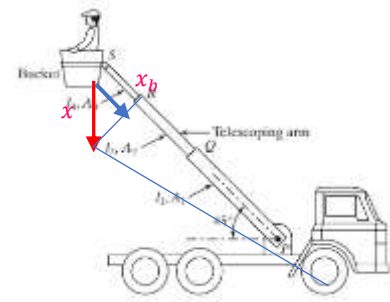
$$\dot{x}_0 = C_2 \omega_n \gg \gg C_2 = \frac{\dot{x}_0}{\omega_n} \quad \left\{ \begin{array}{l} x_0 = C_1 \\ X_0 = \sqrt{x_0^2 + \left(\frac{\dot{x}_0}{\omega_n}\right)^2} \quad \text{and} \quad \varphi = \tan^{-1} \left(\frac{x_0}{\frac{\dot{x}_0}{\omega_n}} \right) \end{array} \right\}$$



$$x(t) = x_0 \cos \omega_n t + \frac{\dot{x}_0}{\omega_n} \sin \omega_n t$$

Example 1

The cockpit of a firetruck is located at the end of a telescopic boom as shown. The cockpit along with the fireman weighs 2000 N . If material of the boom has an elastic modulus $E = 210\text{ GPa}$, $l_1 = l_2 = l_3 = 3\text{ m}$, $A_1 = 20\text{ cm}^2$, $A_2 = 10\text{ cm}^2$, $A_3 = 5\text{ cm}^2$, find the cockpits natural frequency of vibration in the vertical direction. Assume that the mass of the boom is negligible.



When the system vibrates in the vertical direction, let the cockpit along with the fireman moves a distance of x , the axial deformation of the boom be $x_b = x \cos 45$. The same force is transmitted through out the beam along its axis. Hence each section of the boom can be considered as a spring element in series. The axial stiffness of the boom element, $k = \frac{AE}{l}$

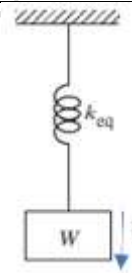
$$k_1 = \frac{A_1 E}{l_1} = \frac{20 \times 10^{-4} \times 210 \times 10^9}{3} = 14 \times 10^7 \frac{\text{N}}{\text{m}}; \quad k_2 = \frac{10 \times 10^{-4} \times 210 \times 10^9}{3} = 7 \times 10^7 \frac{\text{N}}{\text{m}}; \quad k_3 = \frac{5 \times 10^{-4} \times 210 \times 10^9}{3} = 3.5 \times 10^7 \frac{\text{N}}{\text{m}}$$

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} = \frac{1}{14 \times 10^7} + \frac{1}{7 \times 10^7} + \frac{1}{3.5 \times 10^7} \lll k = 2 \times 10^7 \text{ N/m}$$

If the system is replaced by an equivalent spring, the strain energy of the equivalent system, $U_{eq} = \frac{1}{2} k_{eq} x^2$. The strain energy of the fire truck is $U = \frac{1}{2} k x_b^2$. For both system to be equivalent, $\frac{1}{2} k_{eq} x^2 = \frac{1}{2} k x_b^2$

$$\frac{1}{2} k_{eq} x^2 = \frac{1}{2} k (x \cos 45)^2 \lll k_{eq} = 2 \times 10^7 \times \cos^2 45 = 10^7 \text{ N/m}$$

The natural frequency of vibration, $\omega_n = \sqrt{\frac{k_{eq}}{m}} = \sqrt{\frac{2 \times 10^7}{2000/9.81}} = 221.47 \text{ rad/s}$

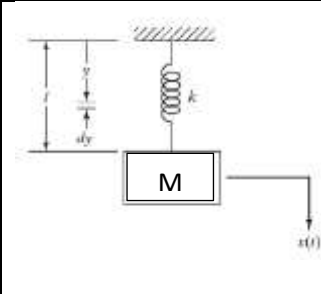
**Example 2**

Determine the effect of the mass of the spring on the natural frequency of the spring-mass system.

Let m be the mass of the spring and l be the length of the spring.

The mass per unit length of the spring $= \frac{m}{l}$

The mass of the infinitesimally small length $dy = \frac{m}{l} dy$



The displacement of the spring at a distance y from the fixed end $= \frac{x}{l} y$

The velocity of the spring element at a distance y from fixed end $= \frac{\dot{x}}{l} y$

Kinetic energy of the spring element $= \frac{1}{2} \left(\frac{m}{l} dy \right) \left(\frac{\dot{x}}{l} y \right)^2 = \frac{1}{2} \frac{m}{l^3} \dot{x}^2 y^2 dy$

Kinetic energy of the total spring, $T = \int_0^l \frac{1}{2} \frac{m}{l^3} \dot{x}^2 y^2 dy = \frac{1}{2} \frac{m}{l^3} \dot{x}^2 \int_0^l y^2 dy = \frac{1}{2} \left(\frac{m}{3} \right) \dot{x}^2$

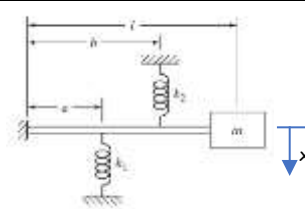
Total kinetic energy of the system, $T_{system} = \frac{1}{2} \left(\frac{m}{3} \right) \dot{x}^2 + \frac{1}{2} M \dot{x}^2$

If the system is replaced by a single equivalent mass having displacement x , the kinetic energy of the equivalent system, $T_{eq} = \frac{1}{2} m_{eq} \dot{x}^2$

$$\frac{1}{2} m_{eq} \dot{x}^2 = \frac{1}{2} \left(\frac{m}{3} \right) \dot{x}^2 + \frac{1}{2} M \dot{x}^2 \lll m_{eq} = M + \left(\frac{m}{3} \right) \ggg \omega_n = \sqrt{\frac{k}{M + \left(\frac{m}{3} \right)}}$$

#Exercise 1

Determine the natural frequency of vibration of the system shown.



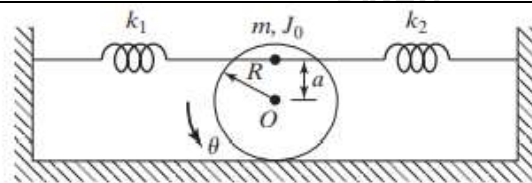
#Exercise 2

An undamped single degree of freedom system consists of a mass 5 kg and a spring of stiffness 2000 N/m. Find the response of the system when the mass is subjected to the following initial conditions. Plot the displacement, velocity, and acceleration in Matlab/Scilab.

- i) $x_0 = 20 \text{ mm}, \dot{x}_0 = 200 \text{ m/s}$ ii) $x_0 = -20 \text{ mm}, \dot{x}_0 = 200 \text{ m/s}$
 ii) $x_0 = 20 \text{ mm}, \dot{x}_0 = -200 \text{ m/s}$ ii) $x_0 = -20 \text{ mm}, \dot{x}_0 = -200 \text{ m/s}$

#Exercise 3

A cylinder of mass m and mass moment of inertia J_0 is free to roll without slipping but is restrained by two springs of stiffnesses k_1 and k_2 as shown. Find its natural frequency of vibration.



#Exercise 4

A steel shaft of 0.05 m diameter and 2 m length is fixed at one end and carries at the other end a steel disc of 1m diameter and 0.1 m thickness. Find the system's natural frequency.

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition